

Clifford Groups and Balanced Sigma–MGFs

Proof, phase correction, and explicit one- and two-qubit verification

Computational verification note

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Abstract

The ordinary one-sided sigma–MGF does not detect unitary-design order. We prove the correct two-sided coefficient statement, isolate a further phase subtlety, and verify the design-detecting coefficient by closing explicit projective Clifford matrix groups in dimensions 2 and 4. The computed frame potentials agree with Haar through degree three and differ at degree four: $(1, 2, 5, 15)$ versus $(1, 2, 5, 14)$ in dimension two, and $(1, 2, 6, 29)$ versus $(1, 2, 6, 24)$ in dimension four.

1 One-sided series and the scalar obstruction

For a finite lift $\tilde{G} \leq U(V)$, put

$$\mathcal{S}_{\tilde{G},V}(\mathbf{t}) = \frac{1}{|\tilde{G}|} \sum_{g \in \tilde{G}} \prod_{i \geq 1} \det(I - t_i g)^{-1}.$$

For a compact group $K \leq U(V)$, define instead

$$\mathcal{S}_{K,V}(\mathbf{t}) = \int_K \prod_{i \geq 1} \det(I - t_i k)^{-1} dk$$

with normalized Haar measure. Expanding each determinant gives

$$[m_\tau] \mathcal{S}_{\tilde{G},V} = \dim \left(\bigotimes_j \mathrm{Sym}^{\tau_j} V \right)^{\tilde{G}}.$$

If the compact group contains the full scalar circle, every positive-degree one-sided coefficient vanishes. Hence

$$\boxed{\mathcal{S}_{K,V} = 1.}$$

for $U(2^n)$ and for the full normalizer $N_{U(2^n)}(P_n)$. A finite scalar lift instead obeys only the congruence imposed by its finite scalar center and may have nonzero lift-dependent coefficients.

2 The two-sided series, correctly phase neutralized

For a specified finite lift $\tilde{G} \leq U(V)$ define

$$\mathcal{B}_{\tilde{G}}(\mathbf{s}, \mathbf{t}) = \frac{1}{|\tilde{G}|} \sum_{g \in \tilde{G}} \prod_i \det(I - s_i g)^{-1} \prod_j \det(I - t_j \bar{g})^{-1}.$$

Character expansion and averaging prove

$$[m_\alpha(\mathbf{s})m_\beta(\mathbf{t})]\mathcal{B}_{\tilde{G}} = \dim \left(\text{Sym}^\alpha V \otimes \text{Sym}^\beta V^* \right)^{\tilde{G}}. \quad (1)$$

There is a useful refinement. Replacing a representative g by zg multiplies the coefficient integrand by $z^{|\alpha|-|\beta|}$. Thus the entire two-sided series is not intrinsically projective. Its *phase-neutral projection*

$$\mathcal{B}_G^0 = \sum_{|\alpha|=|\beta|} [m_\alpha m_\beta] \mathcal{B}_{\tilde{G}} m_\alpha m_\beta \quad (2)$$

is independent of representatives. Equivalently it is the constant term in z after $(\mathbf{s}, \mathbf{t}) \mapsto (z\mathbf{s}, z^{-1}\mathbf{t})$. For the full normalizer, continuous scalar integration performs exactly this projection. All design coefficients used below lie in (2).

3 The design-detecting coefficient

For $(1^k) = (1, \dots, 1)$, one has $\text{Sym}^{(1^k)} V = V^{\otimes k}$. Hence (1) gives

$$[m_{(1^k)}(\mathbf{s})m_{(1^k)}(\mathbf{t})]\mathcal{B}_G^0 = \dim \text{End}_G(V^{\otimes k}) \quad (3)$$

$$= \frac{1}{|G|} \sum_{g \in G} |\text{Tr } g|^{2k}. \quad (4)$$

The last quantity is the k th unitary frame potential.

Let $P_G^{(k)}$ be the average of $g^{\otimes k} \otimes \bar{g}^{\otimes k}$. It is the orthogonal projection onto the corresponding G -fixed space. Since $G \subseteq U(d)$, the Haar fixed space is contained in the G -fixed space. Equality of the coefficient in (4) with the Haar coefficient is equality of the two ranks; the nested spaces, and therefore their orthogonal projections, are equal. This proves

$$\boxed{G \text{ is a unitary } k\text{-design} \iff \Phi_k(G) = \Phi_k(U(d)).}$$

Schur–Weyl duality supplies the exact Haar reference

$$\Phi_k(U(d)) = \sum_{\substack{\lambda \vdash k \\ \ell(\lambda) \leq d}} (f^\lambda)^2,$$

which reduces to $k!$ when $d \geq k$.

4 Independent matrix verification

The program constructs

$$H = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

and, for two qubits, their tensor placements together with CNOT. Matrices are multiplied until closure, with equality taken modulo global phase. The resulting orders are asserted to be 24 and 11,520. No class table or design theorem is used in the numerical average. The Haar comparator is computed independently by the hook-length formula in the Schur–Weyl sum.

n	d	$ \text{Cliff}_n $	k	direct Φ_k	Haar
1	2	24	1	1	1
			2	2	2
			3	5	5
			4	15	14
2	4	11,520	1	1	1
			2	2	2
			3	6	6
			4	29	24

Thus the first total degree at which the canonical phase-neutral coefficient can disagree is four, and the frame coefficient does disagree there. This is precisely the Clifford 3-design/4-design transition established in [1, 2]. The notebook establishes the displayed frame-potential values only for the one- and two-qubit projective groups through $k = 4$. The all-qubit 3-design and failure-of-4-design statements are external theorems supplied by those references.

As a separate regression test, closing the ordinary one-qubit lift generated by H, S gives 192 matrices and scalar center $\mu_8 I$. Its one-sided coefficients for $\tau = (1), (2), (4)$ vanish, while those for (8) and $(4, 4)$ are 1 and 2. This confirms both lift dependence and the scalar selection rule $8 \mid |\tau|$.

5 Reproduction and scope

Run

```
.venv/bin/python for_this_guy/quantum_matrix_groups/
  clifford_groups/verification.py
.venv/bin/marimo edit for_this_guy/quantum_matrix_groups/
  clifford_groups/notebook.py
```

after joining each wrapped command onto one line. The sweep verifies the multiqubit statement for $n = 1, 2$ and $k \leq 4$; the cited theorems supply the all- n result. It makes no inference about odd-prime qudit Clifford groups.

References

- [1] H. Zhu, “Multiqubit Clifford groups are unitary 3-designs,” <https://arxiv.org/abs/1510.02619>.
- [2] H. Zhu, R. Kueng, M. Grassl, and D. Gross, “The Clifford group fails gracefully to be a unitary 4-design,” <https://arxiv.org/abs/1609.08172>.